

Fernando Román Hidalgo

ROBOTICS AND AUTOMATION ENGINEER
41909, Salteras, Sevilla, Spain

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Summary

Engineer with a multidisciplinary profile and systemic vision, featuring professional experience in developing technological solutions and currently completing postgraduate studies. I possess a high capacity for abstraction in solving complex problems and accelerated self-taught learning, enabling me to provide significant value in technically demanding projects. My objective is to consolidate an ambitious career plan, with a focus on automation and embedded software, within dynamic and innovative environments where continuous improvement is the standard.

Experience

Software Engineer

 Airtificial Aerospace & Defense | Seville, Spain

FULLSTACK DEVELOPMENT AND PROJECT MANAGEMENT

April 2025 - Present

- Management and Client Relations:
 - Project management for the development of a process automation application for a major aerospace company.
 - Direct customer interaction for requirements definition and specification.
 - Technical leadership and coordination in the development of custom solutions.
- Technical Development and Automation:
 - Fullstack development, applying coding best practices, efficiency, and organization.
 - Design and implementation of a digital twin for a composite material production plant, oriented towards simulation and optimization of industrial processes.
 - Software integration and deployment using Docker containers.
 - Automation and orchestration of workflows with Apache Airflow.
 - Development of AI models for early failure detection and scheduling algorithms based on CP-SAT.
 - Acquisition and interpretation of data from industrial sensors.

Projects

To see all my projects in detail, check out my Interactive Portfolio.

Honors and Awards

Winner

National Competition Robotics League (LNRC) | Spain

MINISUMO CATEGORY

2024

Education

Master's Degree in Electronic Engineering, Robotics and Automation

 University of Seville
Seville, Spain
2025 - 2026

Bachelor's Degree in Electronic Engineering, Robotics and Mechatronics

 University of Seville
Seville, Spain
2021 - 2025

GPA: 7.5/10

Technological Baccalaureate (Graduated with Honors distinction)
2026-03-18

 IES Heliche
Seville, Spain

Certifications

Advanced Python Course

ONLINE



Jan 2026

Certificate in Advanced English (C1 Advanced)



Cambridge Assessment English

Sep 2024

Skills

Programming	C, C++, Python (OpenCV), MATLAB, Simulink, ROS/ROS2, JavaScript, React, VBA, HTML, CSS, SQL
Electronics	Digital/Analog electronics, microcontrollers (MSP430, ESP32, Arduino...), RTOS, FPGA, automation and control
Simulation	System modeling and simulation (Gazebo, Webots), PID control design, embedded systems integration
DevOps	Docker, Linux, Git, CI/CD, Apache Airflow
Protocols	TCP/IP, MQTT, Zigbee, CAN bus, UART, I2C, SPI
3D Design	Fusion 360, FreeCAD

Soft Skills

Capabilities	Communication, leadership, adaptation, proactivity, critical thinking, punctuality, autonomy, management and teamwork
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Languages

Spanish	Native
English	Professional competence
French	Basic competence